

SAMPLE LESSON CONTENT

Properties of Matter: Grouping Materials Based on Properties

Every material around us has certain properties, or characteristics, that can be observed or tested. These properties include how a material looks and feels, how hard or soft it is, whether it bends easily, and how it behaves when placed in water.

Because materials have different combinations of these properties, they can be grouped, or classified, in many different ways. Grouping materials helps us understand them better and decide which ones are best suited for a particular purpose.

Some Ways to Group Materials by Property

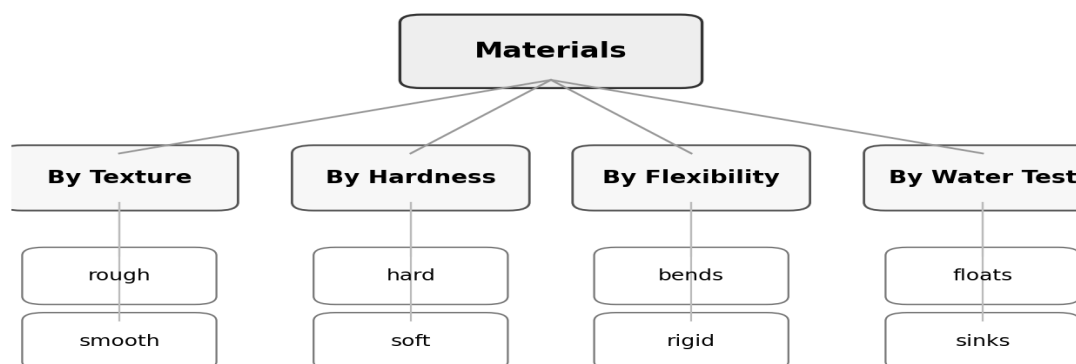


Figure 2. Materials can be grouped in several different ways depending on which property is used as the basis for grouping.

Texture

Texture describes how the surface of a material feels or looks — whether it is rough or smooth. Sandpaper, tree bark, and rocks generally have a rough texture, while glass, silk cloth, and polished metal generally have a smooth texture.

Hardness

Hardness describes how easily a material can be scratched, dented, or reshaped by pressing on it. Materials such as stone, metal, and glass are considered hard because they resist being scratched or reshaped, while materials such as cotton, sponge, and clay are considered soft.

Flexibility

Flexibility describes whether a material can bend without breaking. Rubber bands, cloth, and thin wire are flexible because they can be bent repeatedly. Materials such as dry twigs, glass, and ceramic plates are rigid and tend to snap or crack rather than bend.

Behavior in Water

Materials can also be grouped according to whether they float or sink when placed in water. This generally depends on how dense the material is compared to water. Wood, dry leaves, and most plastics tend to float, while stones, metals, and glass generally sink.

Using Properties to Group Materials

The same set of materials can be grouped in more than one way, depending on which property is being used. For example, a wooden block and a metal spoon could be grouped together as "hard materials," but separated into different groups when tested for floating and sinking, since wood typically floats while metal typically sinks.

Being able to identify and compare the properties of different materials is an important skill, since it allows us to explain why certain materials are chosen for certain uses — such as rubber for tires because of its flexibility, or metal for cooking pots because of its hardness and ability to withstand heat.